**UID EX 10**

**AIM:**

The aim is to create interactive data visualizations, such as line charts and doughnut charts, for a student performance analysis system using JavaScript.

**PROCEDURE:**

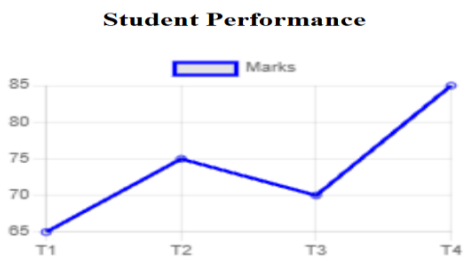
**Step 1:**
Set up the HTML file to create the basic structure of the webpage. Include a canvas element to display the charts.
**Step 2:**
Add required libraries (like Chart.js) using CDN in the HTML file.
**Step 3:**
Create a JavaScript file (script.js) to store student data such as marks, subjects, and performance.
**Step 4:**
Use JavaScript to generate a **line chart** showing marks progression.
**Step 5:**
Create a **doughnut chart** to represent subject-wise performance distribution.
**Step 6:**
Link the JavaScript file to the HTML file and run the program in a browser.

index.html × | # script.css | JS script.js

index.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>Student Performance Charts</title>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
</head>
<body>

    <h2>Student Performance Analysis</h2>

    <!-- Line Chart -->
    <h3>Marks Progress (Line Chart)</h3>
    <canvas id="lineChart" width="400" height="200"></canvas>

    <!-- Doughnut Chart -->
    <h3>Subject-wise Performance (Doughnut Chart)</h3>
    <canvas id="doughnutChart" width="400" height="200"></canvas>

    <script src="script.js"></script>

</body>
</html>
```

```javascript
// Line Chart - Marks Progress
const lineCtx = document.getElementById('lineChart').getContext('2d');

const lineChart = new Chart(lineCtx, {
    type: 'line',
    data: {
        labels: ['Test 1', 'Test 2', 'Test 3', 'Test 4'],
        datasets: [{
            label: 'Marks',
            data: [65, 75, 70, 85],
            borderColor: 'blue',
            fill: false,
            tension: 0.3
        }]
    },
    options: {
        responsive: true
    }
});

// Doughnut Chart - Subject Performance
const doughnutCtx = document.getElementById('doughnutChart').getContext('2d');

const doughnutChart = new Chart(doughnutCtx, {
    type: 'doughnut',
    data: {
        labels: ['Maths', 'Science', 'English', 'Computer'],
        datasets: [{
            label: 'Marks Distribution',
            data: [85, 75, 80, 90],
            backgroundColor: ['red', 'blue', 'green', 'orange']
        }]
    },
    options: {
        responsive: true
    }
});
```

**OUTPUT:**

- A line chart showing student marks over tests
- A doughnut chart showing subject-wise marks distribution